

The empirical recurrence for OEIS A259513: recovering a conjecture that is not written down

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Abstract

OEIS A259513 records “Empirical recurrence of order 56”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 73$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 56, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 56 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A259513 is “Number of $(n+1) \times (6+1)$ 0..1 arrays with each 2×2 subblock having clockwise pattern 0000 0001 0101 0111.”. It has offset 1 and begins

1939, 38409, 784130, 16602297, 355119489, 7648655920, 165145291429, 3570854380319, ...

The entry states:

Empirical recurrence of order 56 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 11:52:11, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 129$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 73$ of the 129, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 73$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 146$ exact terms of the model returns order 56 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{56} c_i a(n-i),$$

c_1	21	c_{15}	−227262085518696	c_{29}	−280033711849485872060	c_{43}	492494446652857
c_2	442	c_{16}	975193049799578	c_{30}	−1663533507680668365030	c_{44}	89437134209094
c_3	−9719	c_{17}	3203706522043071	c_{31}	1397350989212278503514	c_{45}	−34257045073492
c_4	−64564	c_{18}	−16912258034069908	c_{32}	4828178166913366472820	c_{46}	−1319464513234
c_5	1776153	c_{19}	−32337761256685993	c_{33}	−4992992540883877719432	c_{47}	174570815448720
c_6	3467331	c_{20}	218210259936315139	c_{34}	−11001562191858062868816	c_{48}	−1258076814835
c_7	−174511053	c_{21}	224389855017352457	c_{35}	13409526875705255346180	c_{49}	−6277715928527
c_8	57143064	c_{22}	−2143971248517876119	c_{36}	19517990484202804294323	c_{50}	9929640596205
c_9	10534062322	c_{23}	−900665538801617998	c_{37}	−27575656808967036913732	c_{51}	14976008086681
c_{10}	−17866600840	c_{24}	16292452437465195719	c_{38}	−26569463754330922426157	c_{52}	−3375739562854
c_{11}	−420415439157	c_{25}	−338932407139566956	c_{39}	43666070956336652812681	c_{53}	−2111179213468
c_{12}	1120811861857	c_{26}	−96749284245330723085	c_{40}	27052687411323878110060	c_{54}	5864790180096
c_{13}	11598537539864	c_{27}	34353020546372803552	c_{41}	−53129030889859116869470	c_{55}	1320805012517
c_{14}	−40333562825062	c_{28}	451773982269807612985	c_{42}	−19609678602545707394886	c_{56}	−420302777291

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 56, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $56 + 4$ terms removed returns order 56 as well, so $\deg q_{\min} = 56 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $56 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 56 that this sequence satisfies — and that family turns out to have exactly one member.

References

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